

Lab: Explore a Simple Generative Tool

Estimated time needed: 30 minutes

Overview

Generative AI models have revolutionized how you interact with technology, enabling you to create new content, generate realistic images, and translate languages with remarkable accuracy.

In this lab, you will gain hands-on experience with a simple generative AI tool, DataRobot, exploring its capabilities and applications.

Learning Objectives

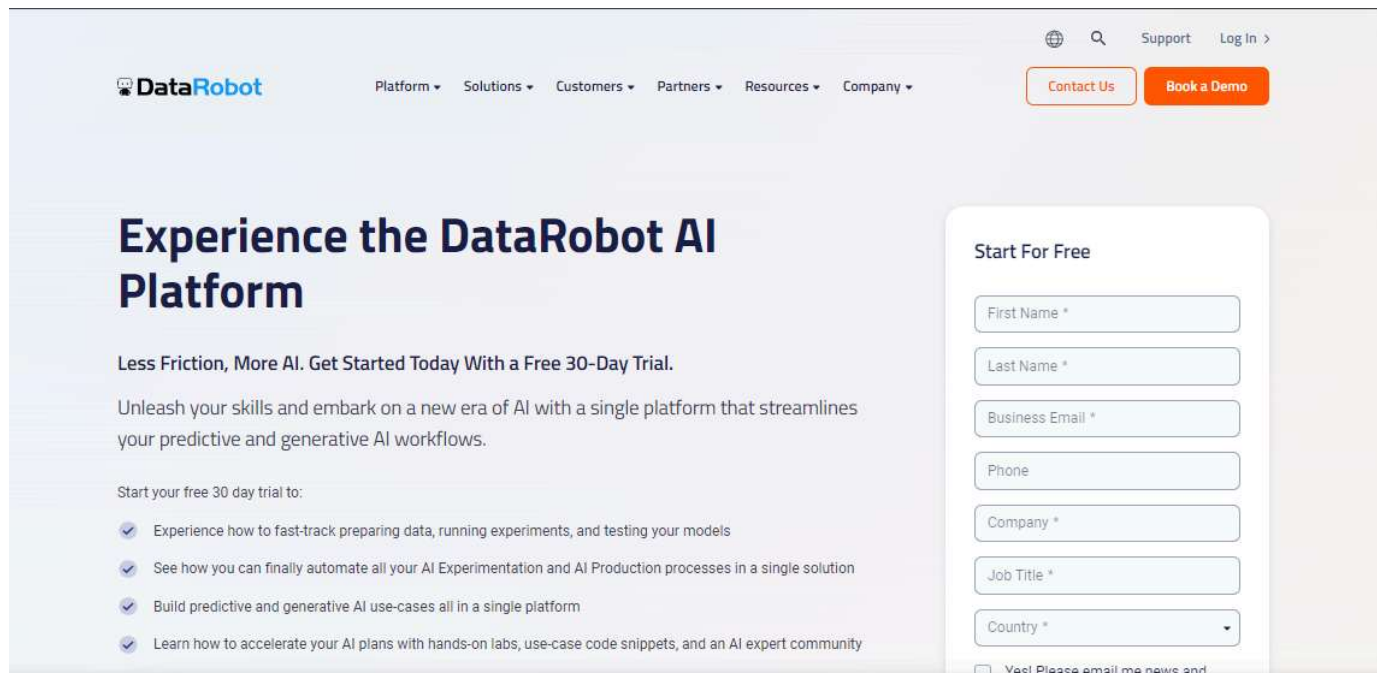
After completing this lab, you will be able to:

- Sign up in DataRobot
- Add a data set to the use case
- Work on model building

Task 1: Sign-up in DataRobot

Step 1: Click www.datarobot.com

Step 2: Fill in the required information under the "Start for free" section and create an account.



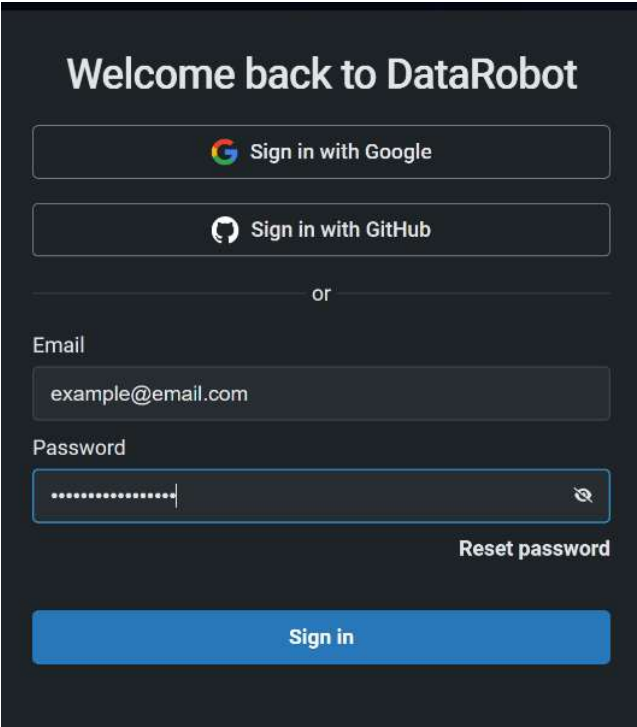
The screenshot shows the DataRobot website's homepage. The header includes the DataRobot logo, navigation links (Platform, Solutions, Customers, Partners, Resources, Company), and links for Support, Log In, Contact Us, and Book a Demo. The main content area features a large heading 'Experience the DataRobot AI Platform' and a subheading 'Less Friction, More AI. Get Started Today With a Free 30-Day Trial.' Below this, a paragraph states: 'Unleash your skills and embark on a new era of AI with a single platform that streamlines your predictive and generative AI workflows.' A section titled 'Start your free 30 day trial to:' lists four benefits with checkmarks. On the right, a 'Start For Free' form is displayed with fields for First Name, Last Name, Business Email, Phone, Company, Job Title, and Country, followed by a checkbox for 'Yes! Please email me news and'.

Note: To access the DataRobot platform, you must sign up using a work email address. If you do not have a relevant work email, an alternative is to create a GitHub account using your Gmail address. Once registered, you can log in to DataRobot using your GitHub credentials.

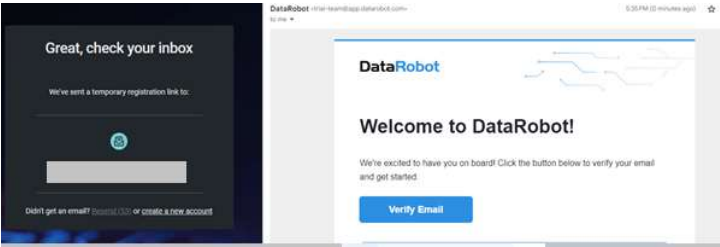
For step-by-step guidance on creating a GitHub account, please refer to the following link:

[GitHub Account Setup Guide](#)

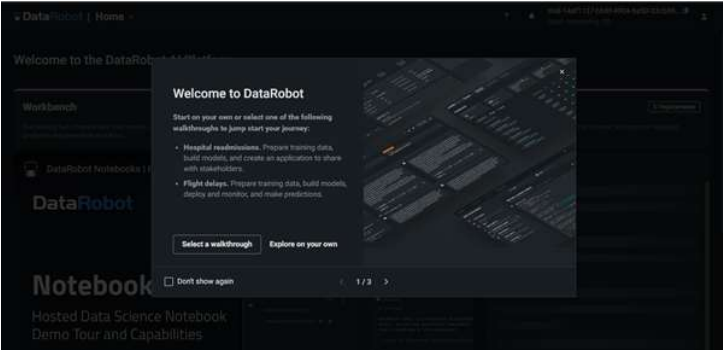
Step 4: A new window will open; select the relevant option for signing up.



Step 5: Confirm your email by clicking **Verify Email** in your inbox.

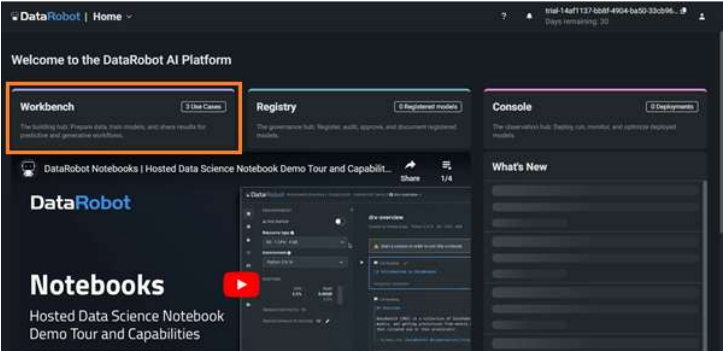


Step 6: Sign up and start your first experience of using the Generative AI tool.
The dashboard will look like the image below. You may like to familiarize yourself with the application by clicking **Select a walkthrough**.

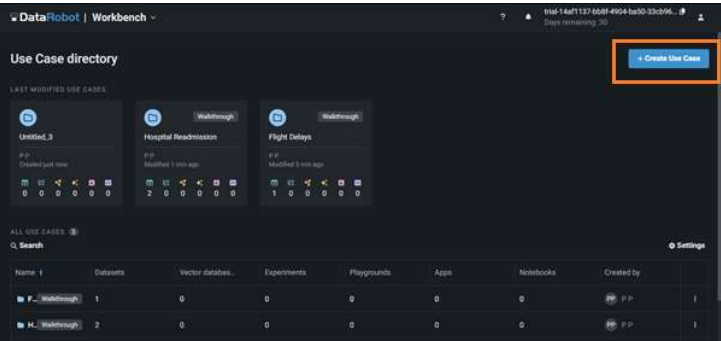


Task 2: Add a data set

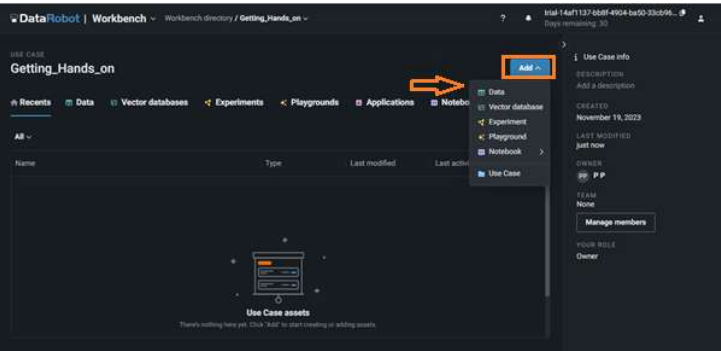
Step 7: The dashboard will appear shortly, and your screen will look as shown below. Click **Workbench**.



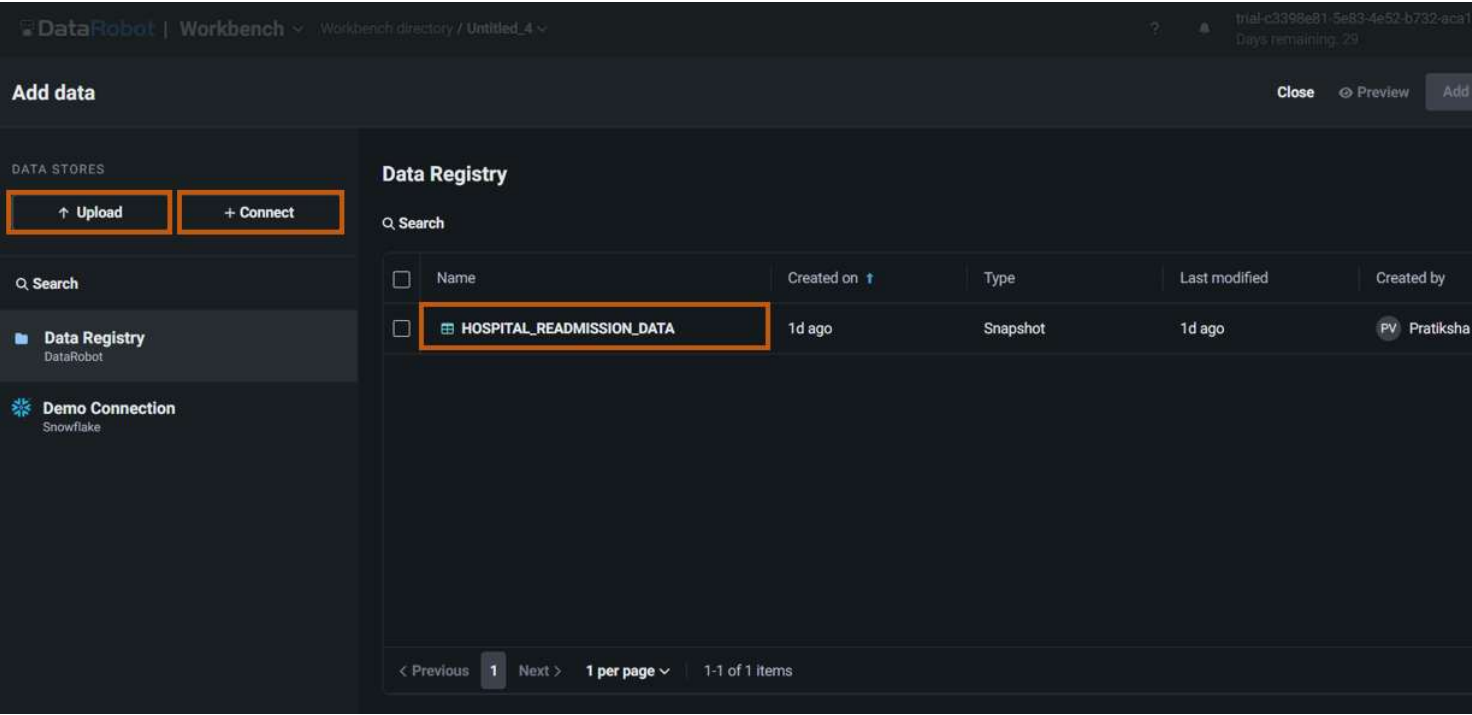
Step 8: Click **Create Use Case**.



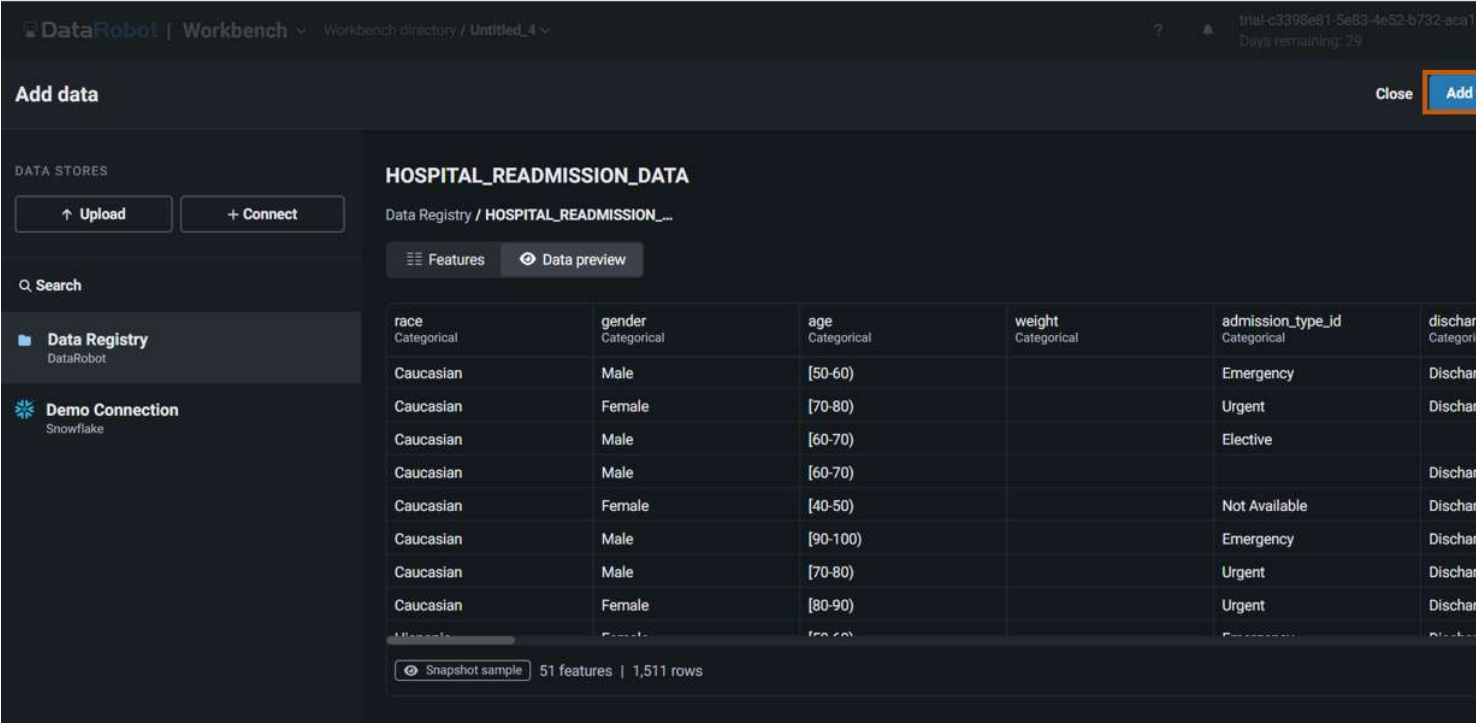
Step 9: Click **Add** and **Data** to include the data set in your use case.



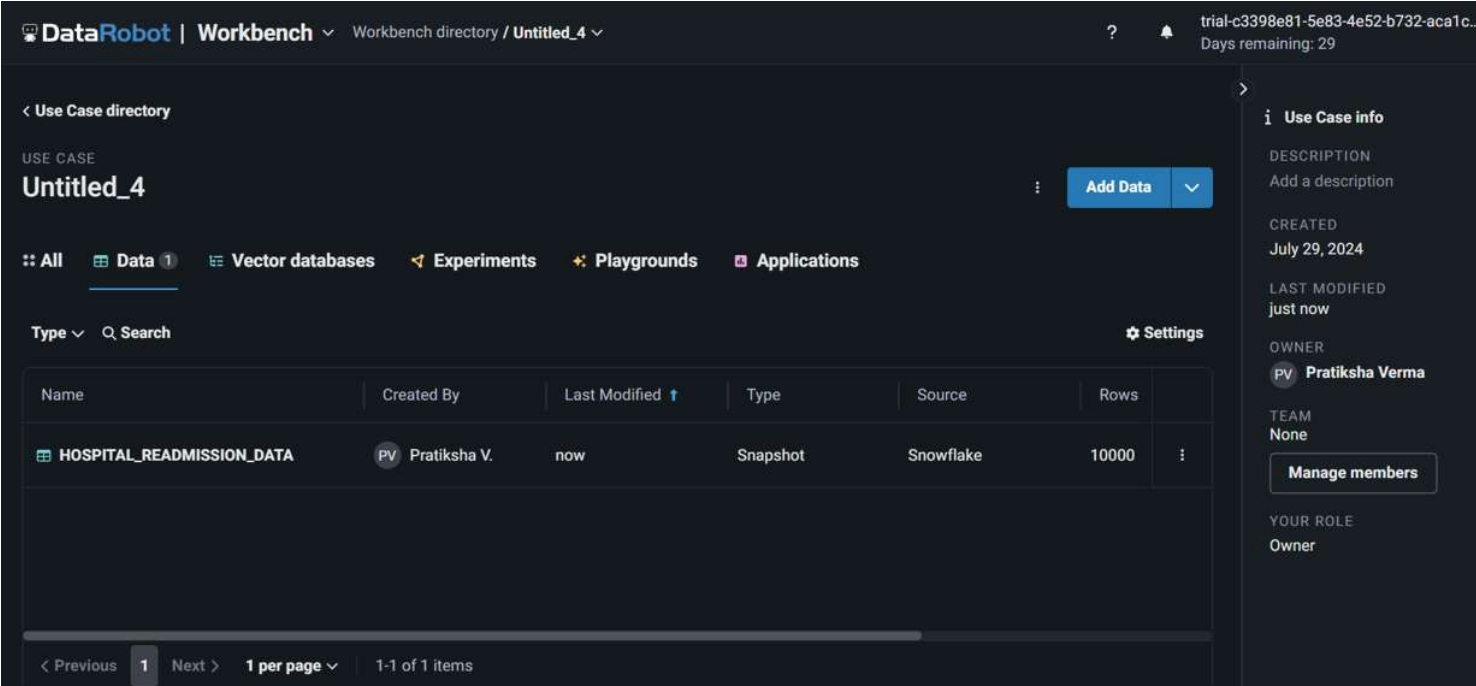
Step 10: **Upload** your data set or **Connect** to the data source; however, for this lab, you can select an in-built sample data set *HOSPITAL_READMISSION_DATA*.



Step 11: Once you select the data set, you can see a preview of it. You can also view the data set's features, as shown below. Click **Add to Use Case**.



Step 12: After you add the data set to the use case, the workbench will appear as shown below. You can click the data set to see the feature insights.



Step 13: Explore the All Features menu to display specific features.

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA** Jul 29th, 2024 10:39 AM Snapshot Data actions

HOSPITAL_READMISSION_DATA

Data preview **Features** Feature lists

Show insights Show features from: All Features + Create feature list

Search

DATAROBOT FEATURE LISTS

- All Features 51
- Informative Features 40
- Raw Features 51

race	gender	age	weight	admission_type_id	discharge_status
Caucasian	Male	[70-80]	==Missing==	Emergency	Discharged
AfricanAmerican	Male	[60-70]	[75-100]	Urgent	Discharged
Other	Female	Other	Other	Other	Other

Snapshot sample 51 features | 1,511 rows

Information

DATE ADDED: July 29, 2024

ADDED BY: Pratiksha Verma

FEATURES: 51

ROWS: 10,000

SIZE: 5.48 MB

DATABASE: Snowflake / Demo Connect

SCHEMA: TRIAL_READONLY

TABLE: HOSPITAL_READMISSION_DATA

Dataset Versions

Task 3: Work on Data Modeling

Step 14: Click **Start**. You will have options **Modelling** and **Start wrangling**. You can try data wrangling if you want to. For this lab, you will work on model building. Click **Start** and select **Modelling**. It will take a while to prepare a data set for modelling.

DataRobot | Workbench Workbench directory / Untitled_4 / **HOSPITAL_READMISSION_DATA** Jul 29th, 2024 10:39 AM Snapshot Data actions

HOSPITAL_READMISSION_DATA

Data preview Features Feature lists

Show insights Show features from: All Features + Create feature list

Start wrangling

Start modeling

Start feature discovery

Download dataset

Remove dataset

race	gender	age	weight	admission_type_id	discharge_status
Caucasian	Female	[70-80]	==Missing==	Emergency	Discharged
AfricanAmerican	Male	[60-70]	[75-100]	Urgent	Discharged
Other	Female	Other	Other	Other	Other

Snapshot sample 51 features | 1,511 rows

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Dataset Versions

Step 15: Once done, you need to select the **Target feature**. Select **readmitted** as your target feature.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c Days remaining: 29

Set up new experiment

Dataset Target Additional settings Exit < Back

Target feature
Select the feature to make predictions on.

payor_code
pioglitazone
race
readmitted
readmission
troglitazone
tolbutamide
tolazamide
time_in_hospital

	Uniq...	Missi...	Mean	Std Dev
readmitted	7	9592	-	-
troglitazone	35	Categorical	1	0
tolbutamide	30	Categorical	2	0
tolazamide	36	Categorical	2	0
time_in_hospital	7	Numeric	14	0

Experiment summary
HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset
Name HOSPITAL_READMISSION_DATA
Rows 10,000
Features 51

Target
No target selected

Step 16: The workbench screen will be displayed as shown below. Click **Next**.

DataRobot | Workbench Workbench directory / Untitled_4

trial-c3398e81-5e83-4e52-b732-aca1c Days remaining: 29

Set up new experiment

Dataset Target Additional settings Exit < Back

Target feature
Select the feature to make predictions on.

readmitted

Target type: Binary classification ⓘ
Positive class: ☐ 0 ☒ 1 ⓘ

Modeling mode
Set the mode used for selecting which blueprints to build when training models.
Quick Autopilot

Optimization metric
Set the metric used when training models to evaluate and optimize accuracy.
LogLoss (Accuracy) Recommended

Experiment summary
HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset
Name HOSPITAL_READMISSION_DATA
Rows 10,000
Features 51

Target
Feature readmitted
Target type Binary classification
Positive class 1
Modeling mode Quick Autopilot
Optimization metric LogLoss
Training feature list Informative Features

Partitioning

Value	Number of rows
False	~5800
True	~3800

Step 17: You can modify the model setting in **Additional Settings**; once done, click **Next** and then click **Start modelling**.

DataRobot | Workbench Workbench directory / Untitled_4

Set up new experiment Dataset Target Additional settings Exit < Back Start

Data partitioning Time series modeling Preview Additional settings

Partitioning method
Select the method for assigning rows to partitions when training models.

Stratified sampling
Rows are assigned to ensure similar target distribution across each partition.

Validation type

- ☒ **Cross-validation**
Trains models on a specified number of folds, maximizing data use but also increasing run time.
- ☐ **Training-validation-holdout**
Splits data into three partitions: trains models on the training set, assess performance on the validation set, and evaluates the model on unseen data in the holdout set.

Cross-validation folds
Enter a value from 2 - 50.

Holdout percentage
Set the subset of data that is unavailable during training and validation. Enter a value

Experiment summary
HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

Dataset	
Name	HOSPITAL_READMISSION_DATA
Rows	10,000
Features	51

Target	
Feature	readmitted
Target type	Binary classification
Positive class	1
Modeling mode	Quick Autopilot
Optimization metric	LogLoss
Training feature list	Informative Features

Partitioning

Step 18: Building models will take a while.

DataRobot | Workbench Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57

trial-c3398e81-5e83-4e52-b732-aca1c... Days remaining: 29

Experiment Comparison

View experiment info Filter Validation: LogLoss Search

MODELS 0/0

Models are building...

Keras Slim Residual Neural Network Classifier using... Building...

Informative Features
64% (6,400 rows)

Elastic-Net Classifier (L2 / Binomial Deviance) Building...

Informative Features
64% (6,400 rows)

Step 19: once the modelling is complete, you can pick a model of your choice, and the DataRobot will show the **Model Overview**.

The screenshot shows the DataRobot Workbench interface. The top navigation bar includes 'DataRobot | Workbench', a breadcrumb 'Workbench directory / Untitled_4 / HOSPITAL_READMISSION_DATA - 2024-07-30 11:23:57', and a trial status 'trial-c3398e81-5e83-4e52-b732-aca1' with 'Days remaining: 29'. The left sidebar has tabs for 'Experiment' and 'Comparison', with 'Experiment' selected. Below the sidebar, the 'Model Overview' page is displayed for an 'Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance)'. The page is divided into several sections: 'Training scores: LogLoss' (Validation: 0.6089, Cross-validation: 0.6111, Holdout: 0.6111), 'Training settings' (Training feature list, Informative Features: 64% (6,400 rows), Training sample size), 'Blueprint', 'Feature Impact', 'Feature Effects', 'Individual Prediction Explanations', and 'ROC Curve'. The 'Blueprint' section is highlighted with an orange box. The 'Feature Impact' section is also highlighted with an orange box.

Step 20: You can explore various model overview components like **Blueprint**, **Feature Impact**, and so on.

The screenshot shows the DataRobot Workbench interface with the 'Blueprint' and 'Feature Impact' sections highlighted. The 'Blueprint' section displays a flowchart of the model's data processing pipeline: Data → Categorical Variables → One Hot Encoding → Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance) → Prediction. The 'Feature Impact' section shows a horizontal bar chart of feature importance for the Elastic-Net Classifier. The chart lists features and their corresponding importance scores, sorted in descending order. The top features are 'number_inpatient', 'number_diagnoses', 'medical_specialty', 'admission_type_id', 'discharge_disposition_id', 'admission_source_id', 'diag_3', 'diag_1', 'age', 'diag_2', 'num_lab_procedures', 'number_emergency', 'diabetesMed', 'race', 'insulin', 'payer_code', 'change', 'diag_2_desc', and 'prioritization'. The 'number_inpatient' feature has the highest importance, followed by 'number_diagnoses' and 'medical_specialty'.

Step 21: If you have test or unseen data, you can also make predictions by clicking **Make Predictions** under **Model actions**.

Model Overview
Elastic-Net Classifier (mixing alpha=0.5 / Binomial Deviance) ☆

Training scores: LogLoss
Validation: 0.6089
Cross-validation: 0.6111
Holdout: 0.6111

Training settings
Training feature list: Informative Features
Training sample size: 64% (6,400 rows)

Model actions
Register model
Make predictions
Create app
Generate compliance report
Delete model

Make Predictions
Make new predictions
Prediction dataset: Prediction datasets cannot exceed 5.00 GB.
Drop file(s) here to upload or Choose file
Upload local file
Use model training data
Data registry
Prediction options
Include additional feature values in prediction dataset
Add all features
Add specified features
All features selected
Include Prediction Explanations

Download recent predictions
Predictions are stored and available to download for 48 hours.
No recent predictions

Step 22: You can also click **Generate compliance report** and **download compliance report** for your use case.

Model actions
Register model
Make predictions
Create app
Generate compliance report
Delete model

Table of Contents

- 1 How To Use This Document
- 2 DataRobot Model Development Documentation
- 3 Executive Summary and Model Overview
 - 3.1 Model Stakeholders
 - 3.2 Model Development Purpose and Intended Use
 - 3.3 Model Description and Overview
 - 3.4 Overview of Model Results
 - 3.5 Model Interdependencies
- 4 Model Data Overview
 - 4.1 Feature Association
 - 4.2 Data Source Overview and Appropriateness
 - 4.3 Input Data Extraction, Preparation, and Quality & Completeness

Conclusion

In this lab, you have signed up in DataRobot, added a data set in a use case, and worked on data modelling.

Author(s)

[Dr. Pooja](#)



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